AI201 Artificial Intelligence - Syllabus

Module 1: Introduction to AI

Artificial intelligence is the study of agents that perceive their environment and take actions to achieve goals. The term artificial intelligence was coined by John McCarthy at the Dartmouth workshop in 1956. Alan Turing proposed the Turing test in 1950: a machine passes if a human judge cannot tell its answers from a person's. A rational agent chooses the action expected to maximise its performance measure. Agent types are simple reflex, model-based reflex, goal-based, utility-based and learning agents. A task environment is described using PEAS: Performance measure, Environment, Actuators and Sensors. Environments are classified as fully or partially observable, deterministic or stochastic, episodic or sequential, static or dynamic, discrete or continuous, and single-agent or multi-agent. Reference text: Russell and Norvig, Artificial Intelligence: A Modern Approach.

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Module 2: Problem Solving and Search

Problem solving is modelled as search through a state space from an initial state to a goal state. Breadth-first search expands the shallowest node first; it is complete and optimal when every step costs the same, but its time and memory grow as O(b^d). Depth-first search uses memory O(bm) but can get stuck in infinite paths. Uniform-cost search expands the node with the lowest path cost g(n). Iterative deepening combines the memory use of depth-first search with the completeness of breadth-first search. A* search ranks nodes by the evaluation function f(n) = g(n) + h(n), where h(n) estimates the remaining cost. A heuristic is admissible if it never overestimates the true cost to reach the goal; with an admissible heuristic A* is optimal. Greedy best-first search uses only h(n). Hill climbing can stop at a local maximum; simulated annealing escapes by sometimes accepting worse moves. Minimax search with alpha-beta pruning is used for two-player games.

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Module 3: Knowledge Representation

A knowledge base is a set of sentences in a formal language that an agent can reason over. Propositional logic uses symbols and connectives such as AND, OR, NOT and implies. First-order logic adds objects, predicates and the quantifiers for all and there exists. Modus ponens is an inference rule: from P and the sentence P implies Q, infer Q. Resolution is a complete inference method for clauses in conjunctive normal form. Forward chaining starts from known facts and derives new ones; backward chaining starts from a goal and works back to the facts. A Horn clause has at most one positive literal, and Prolog programs are built from Horn clauses. Semantic networks and frames organise knowledge as nodes and slots. An ontology defines the concepts of a domain and the relations between them, and is used by the semantic web.